

IEEE Standards and Their Power Carrying Capabilities

Table 1

Name	IEEE Standard	Power to Powered Device (PD)	Max. Power per Port	Energised Pairs	Supported Applications and Supported Devices
PoE	IEEE 802.3af	12.95W	15.4W	2-pair	Static surveillance cameras, VoIP phones, wireless access points
PoE+	IEEE 802.3at	25.5W	30W	2-pair	PTZ cameras, video IP phones, alarm systems
PoE++	IEEE 802.3bt (Type 3)	51W	60W	4-pair	Video conferencing equipment, multi-radio wireless access points
PoE++	IEEE 802.3bt (Type 4)	71.3W	100W	4-pair	Laptops, flat screens

Power at Source and Power Delivered at the Node

Table 2

	Type 1 802.3af			Type 2 802.3at	Type 3 802.3bt		Type 4 802.3bt	
Power Class	Class 1	Class 2	Class 3	Class 4	Class 5	Class 6	Class 7	Class 8
Power from PSE	4W	7W	15.4W	30W	45W	60W	75W	90W
Power delivered to PD	3.84W	6.49W	13W	25.5W	40W	51W	62W	71.3W

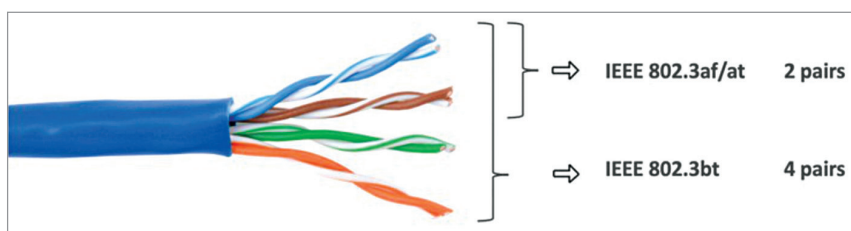


Fig. 9: Cable pairs used for two different PoE standards

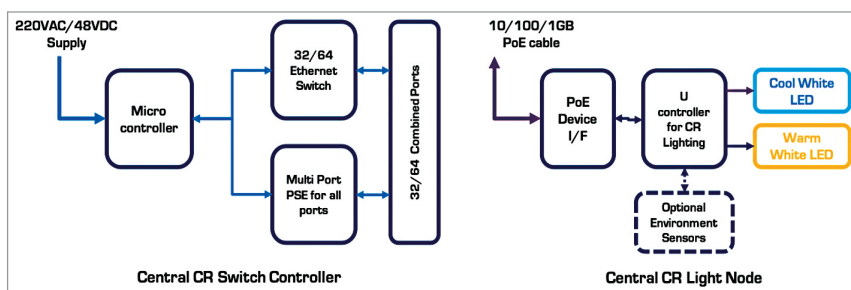


Fig. 10: A basic PoE lighting system

Table 2 shows the actual power at the source and power delivered at the node for better understanding of the real capabilities of different variations of the standards.

Functional blocks

With power handling capability of PoE ranging from 25W to 70W pretty much all the luminaires used in homes and offices can be easily implemented using PoE. A basic PoE lighting system has two major blocks, which are shown in Fig. 10.

The first block is of central con-

troller that performs following three functions:

1. Acts as a central controller for controlling all the human-centric/circadian rhythm system in the building.
2. Acts as a PoE source node along with an Ethernet switch to feed the power to the lights. The PoE powered human-centric/circadian rhythm switch can work with off-the-shelf Ethernet switches and associated components for 32 or 64 ports.
3. Supplies power through the Ethernet cable (up to 100 metres maximum) for the lights to work.

The second block is of circadian rhythm light node that has PoE device interface chip with built-in DC/DC converter. It is managed by the micro-controller which receives commands from the PoE switch controller. Based on the commands it controls the warm and cool LED based lights to produce the right amount of stimulus based on programming, time of day, and location.

More complex systems can be equipped with environmental sensors including ambient light sensor to ensure correct amount of light is configured by correlating the internal lighting with the measured value and external light coming through the windows.

Advantages of PoE powered lighting

PoE is basically a power delivery mechanism using the existing Ethernet wiring for up to 30W end node consumption. For higher power levels CAT6E or equivalent are preferred. Some key advantages are:

1. Separate power and control wiring are not needed in building which uses 20W troffer lamps (typical 2x2 false-ceiling-mounted square luminaires consuming 20 to 30W that are normally used in commercial environments).

2. Since the power and the control are through the same cable, and normally most offices have wired building, implementing a human-centric/circadian rhythm system can be managed in the existing network environment.

3. Since the lighting system has become part of the network, separate back-up for the lighting is not needed. Typically, office complexes have inverter as a backup for the lights till the generators take over. In the case of PoE based lighting this issue is taken care of as the network equipment is backed up through UPS and generators.

4. With the central controls, lighting can be programmed through human-centric/circadian rhythm system and can also be used to optimise the lighting requirement based on the active presence of humans in the room. **EFY**



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